

## Assignment 5

### 1 Requirements

You are expected to complete all exercises. For submission, please put all your answers in a PDF file and submit it. The name of the file should follow the format "student-ID\_assignmentNumber" (e.g., 2024111054\_assignment5). **The submission deadline is 11:59 PM, Nov 24.**

### 2 Required Exercises (100 points)

**Exercise 1:** For the SDD in Figure 1, give annotated parse trees for the following expressions:

1.  $(3 + 4) * (5 + 6)n$  [20points]
2.  $1 * 2 * 3 * (4 + 5)n$  [20 points]
3.  $(9 + 8 * (7 + 6) + 5) * 4n$  [20 points]

|    | PRODUCTION                     | SEMANTIC RULES                  |
|----|--------------------------------|---------------------------------|
| 1) | $L \rightarrow E \mathbf{n}$   | $L.val = E.val$                 |
| 2) | $E \rightarrow E_1 + T$        | $E.val = E_1.val + T.val$       |
| 3) | $E \rightarrow T$              | $E.val = T.val$                 |
| 4) | $T \rightarrow T_1 * F$        | $T.val = T_1.val \times F.val$  |
| 5) | $T \rightarrow F$              | $T.val = F.val$                 |
| 6) | $F \rightarrow ( E )$          | $F.val = E.val$                 |
| 7) | $F \rightarrow \mathbf{digit}$ | $F.val = \mathbf{digit.lexval}$ |

Figure 1: Syntax-directed definition of a simple desk calculator

**Exercise 2:** What are all the topological sorts for the dependency graph of Figure 2? One sort mentioned during lecture is 1,2,3,...,9 (slide #17 of Chapter 4). [20 points]

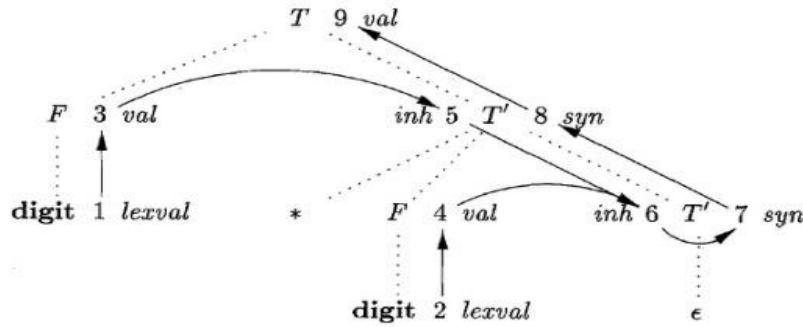


Figure 2: A dependency graph

**Exercise 3:** Below is a grammar for expressions involving operator `+` and integer or floating-point operands. Floating-point numbers are distinguished by having a decimal point. Give an SDD to determine the type of each term `T` and expression `E`. [20 points]

$$E \rightarrow E + T \mid T$$

$$T \rightarrow num \cdot num \mid num$$

